
CE 591: Special Topics in Civil and Mechanical Engineering

Course Code	CE 591
Course Name	Special Topics in Civil and Mechanical Engineering
L-T-P-C	3-0-0-3
Prerequisites	Solid Mechanics, Higher Engineering Mathematics
Intended for	UG (4th year Civil and Mechanical Engineering), PG (School of Engineering)
Distribution	Elective

Course Contents

- **Module I:** Physical problems. Mathematical models. Ordinary and partial differential equations. Elliptic, parabolic and hyperbolic equations. Boundary, initial and boundary-initial value problems. [5 Lectures]
- **Module II:** Some analytical solutions for simple idealized problems. Difficulties in obtaining analytical solutions. Numerical solutions. Finite difference methods for partial differential equations. [13 Lectures]
- **Module III:** History of finite element method for structural problems. Finite element method for non structural problems. Finite element method for partial differential equations. [20 Lectures]
- **Module IV:** Finite element software development for solution of some practical problems. [4 Lectures]

Text Books

1. O.C. Zienkiewicz and K. Morgan, **Finite Elements and Approximation**, Dover Publications, 1983.
2. S.C. Chapra and R.P. Canale, **Numerical Methods for Engineers: With Programming and Software Application**, WCB/McGraw-Hill, 1998.

References

1. D.V. Hutton, **Fundamentals of Finite Element Analysis**, McGraw Hill Education, 2017.
2. S.S. Rao, **The Finite Element Method in Engineering**, 5th edition, Elsevier, 2011.